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Climate change and infectious diseases

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Climate change will alter the distribution and burden of infectious diseases. Anticipating future impacts requires characterizing how climate drivers alter transmission of vector-borne, waterborne and respiratory pathogens, accounting for nonlinear relationships between climate variables and disease outcomes. Here we show how inference from laboratory and observational studies in the present can be used to develop projections for the future impact of climate change on infectious disease, and to understand how climate change to date may have impacted existing disease trajectories. We synthesize data from multiple pathogens to show the broad implications of climate change for spatial and temporal outbreak patterns and predictability. One of the most immediate consequences of climate change may be to exacerbate the impact of weather extremes and climate variability, requiring novel data streams and modeling tools to tune interventions. At the same time, climate change is set to occur against the backdrop of demographic change; therefore, determining global shifts in vulnerability, both in the present and the future, is an important task for public health.

As the climate changes, it has the potential to alter the infectious disease burden globally[1]. In the decades leading up to 2020, expectations for the future burden of disease were often framed in the context of 'the epidemiological transition'[2]. This conceptual model posits a declining importance of infectious diseases globally in pace with economic growth, alongside a growing importance of chronic diseases. However, the COVID-19 pandemic, estimated to have killed over 27 million people globally[3], illustrated the continuing dangers of infectious disease in general, and the potential devastation that can be wrought by directly transmitted infections in particular. Vector-borne diseases are also rife, with malaria continuing to kill more than half a million people every year[4], and dengue expanding around the world[5,6]. Sanitation is improving globally, yet devastating outbreaks of waterborne diseases like cholera continue to occur[7]. Many of these patterns of disease outbreaks are contingent on climate impacts. Bounding expectations for how infectious disease will be altered in a changing climate is thus an important question for public health planning.

In order for researchers to understand the future risk of climate change on infectious disease, they must first characterize how climate variables affect disease transmission and incidence in the present. In this Review, we consider key advances across different spatial and temporal scales: from the pathogen scale in the present, to the population scale in the future. We begin by considering how climate variables, such as temperature, humidity and precipitation, affect pathogens at the point of transmission. We then consider how these effects on transmission impact the present-day seasonality and range of infectious disease outbreaks in populations. We show how this understanding of present-day dynamics can be leveraged to understand the future impact of climate change, as well as the effect of climate change to date. Finally, we consider two emerging areas of research: the effect of weather extremes and climate variability on infectious disease outbreaks, and how demographic change may intersect with climate change to alter the risk from infectious disease in the future.

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We focus on three main pathogen groups: vector-borne diseases, directly transmitted diseases (including respiratory pathogens) and diseases with an environmental reservoir. In other pathogen groups—such as sexually transmitted pathogens and foodborne pathogens—evidence indicating that climate affects transmission is limited, and we do not discuss this here. Zoonotic disease emergence, as well as infections in plants, wildlife and livestock[1], may also be modulated by a shifting climate[1,8], but our primary focus is on currently circulating infectious diseases of humans.

The role of climate variables in disease transmission

Understanding how climate change may affect infectious diseases requires understanding the role of climate variables in disease transmission. Studies linking climate variables to disease transmission derive evidence from both laboratory and observational studies. The pathways linking environmental drivers to disease transmission, and the types of study required to elucidate these pathways, depend on both pathogen type and the climate variable. For a large proportion of the pathogen groups that are the focus of this Review, there has been work done to understand the effect of temperature on transmission. Recently, the role of humidity—both in terms of absolute humidity (the mass of water molecules per unit volume of air) and relative humidity (the mass of water molecules relative to the maximum at the same temperature)—has been the focus of emerging research[9]. There has been far less work to understand the effect of precipitation on disease transmission; challenges with experimental setups and remote-sensing data have limited work in this area to date.

Vector-borne diseases

For pathogens transmitted by a vector (for example, mosquitoes, ticks and fleas), changes in temperature, humidity or other climate parameters can alter the rate of pathogen development and/or the reproduction and behavior of the vector (Fig. 1). A mixture of experimental and observational work shows a strong effect of temperature on mosquito[10], tick[11] and flea[12] traits such as fecundity, with implications for dengue and malaria, Lyme disease and plague, respectively (among many other diseases). Increasing temperature is also associated with an increase in *Aedes* mosquito development rate[13] and biting rate[14], which may explain observations that warming is associated with increases in dengue infections in much of the Americas and Asia[5]. The direction and magnitude of changes depend on the pathogen and vector in question, and biological limits mean that many relationships are nonlinear. Optimum temperatures for malaria transmission are likely substantially lower than for dengue[15,16], such that warming in areas where temperatures are already above the optimum for malaria may see a transition from malaria to dengue burden[17]. Much of our inference has been rooted in either vector performance studies in controlled temperature lab settings, or observational studies. More recent work suggests the role of humidity has been overlooked in both laboratory and observational studies on malaria transmission[9,18]. Further studies linking epidemiological patterns in vector-borne diseases to climate variables are needed, given the difficulty in inferring limits to real-world vector performance from experiments with vectors grown in stable lab conditions[19].

Directly transmitted diseases

While climate variables are widely recognized as influencing vector-borne disease transmission, growing experimental and epidemiological evidence indicates that directly transmitted diseases are also climate sensitive[10]. Some of the most convincing evidence for this comes from animal models; however, as with vector-borne diseases, this evidence may only hint at the potential mechanistic drivers at the population scale. Lowen et al.[20] found that the spread of influenza in aerosols between infected and susceptible guinea pigs was sensitive to temperature and relative humidity, although later work suggests absolute humidity may be the best predictor[21,22]. Mechanistically, climate variables may modulate the degree of susceptibility of uninfected hosts (a phenomenon with relevance across all pathogen groups discussed here), pathogen viability and the efficacy of transmission between hosts.

With regard to host susceptibility, there is experimental evidence—primarily from in vitro and ex vivo experiments and animal models—that humidity and/or temperature modulate the integrity and function of the airway epithelium[23] as well as innate and adaptive[23–25] immune responses to respiratory viral infections. In terms of pathogen viability, generally speaking, when respiratory particles are emitted, evaporation causes them to quickly lose heat and water to equilibrate with their surroundings[26], resulting in rapid changes in solute concentration, pH and temperature[27]. Recently, Morris et al.[28] proposed a model in which temperature inactivates viruses following an Arrhenius equation[29], and decreasing relative humidity decreases virus survival only above a critical threshold value, by concentrating solutes; below this threshold, solute crystals are formed and inactivation is no longer sensitive to relative humidity. These results indicate a ‘U-shaped’ effect of humidity on transmission that is consistent with predictions based on epidemiological data for respiratory syncytial virus (RSV)[30] and influenza[31]. It is critical, however, to use accurate models for respiratory fluids when performing these experiments, as the presence of proteins and mucus[32] has been shown experimentally to impact virus survival, particularly at moderate relative humidity. Moving forward, careful and systematic experiments evaluating virus survival in relevant fluids will be key for establishing functional relationships between pathogen survival and climate variables.

Finally, in terms of transmission efficacy, the seasonality of directly transmitted diseases is also impacted by the social behavior of hosts. Recent efforts to quantify this proposed a geographically and temporally resolved metric for indoor activity, obtained from anonymized mobile GPS panel data in the context of respiratory diseases[33]. Increased indoor activity may impact transmission efficacy for respiratory pathogens by changing airborne contact times between susceptible/infected individuals, and modifying the extent of different transmission modes (that is, droplet contact from close proximity conversations). Increased indoor activity may also feed back into modifying host susceptibility and pathogen survival, owing to the nature of indoor versus outdoor ambient air conditions (for example, lower humidity)[33].

Pathogens transmitted through an environmental reservoir

This group includes waterborne and soilborne diseases such as cholera and coccidioidomycosis. Changes in temperature and precipitation impact pathogen replication and survival in the reservoir, and human exposure through direct and indirect mechanisms. Directly, the abundances of *Vibrio* spp.[34] and *Salmonella Typhi*[35] in water are expected to increase with temperature. Heavy rainfall can increase waterborne disease transmission as flooding contaminates wells and surface water[36,37] and may increase exposure to free-living pathogens, as observed for leptospirosis[38]. The effect of rainfall on disease spread is exacerbated by increased flood risk in coastal areas experiencing sea level rise, storm surge and the disruption of water treatment systems[39]. By damaging infrastructure and affecting reservoir conditions (for example, altering water salinity to favor *Vibrio* spp. replication[40]), extreme events such as hurricanes can have cumulative effects on waterborne disease spread. However, waterborne disease transmission may also be enhanced under low precipitation (drought) conditions because bacteria are more concentrated in the limited water supply[41], and contaminated water is more likely to be consumed when water is scarce[42]. The nonlinear relationship between precipitation and transmission suggests that waterborne diseases will spread more easily as climate variability increases and extreme weather events become more frequent.

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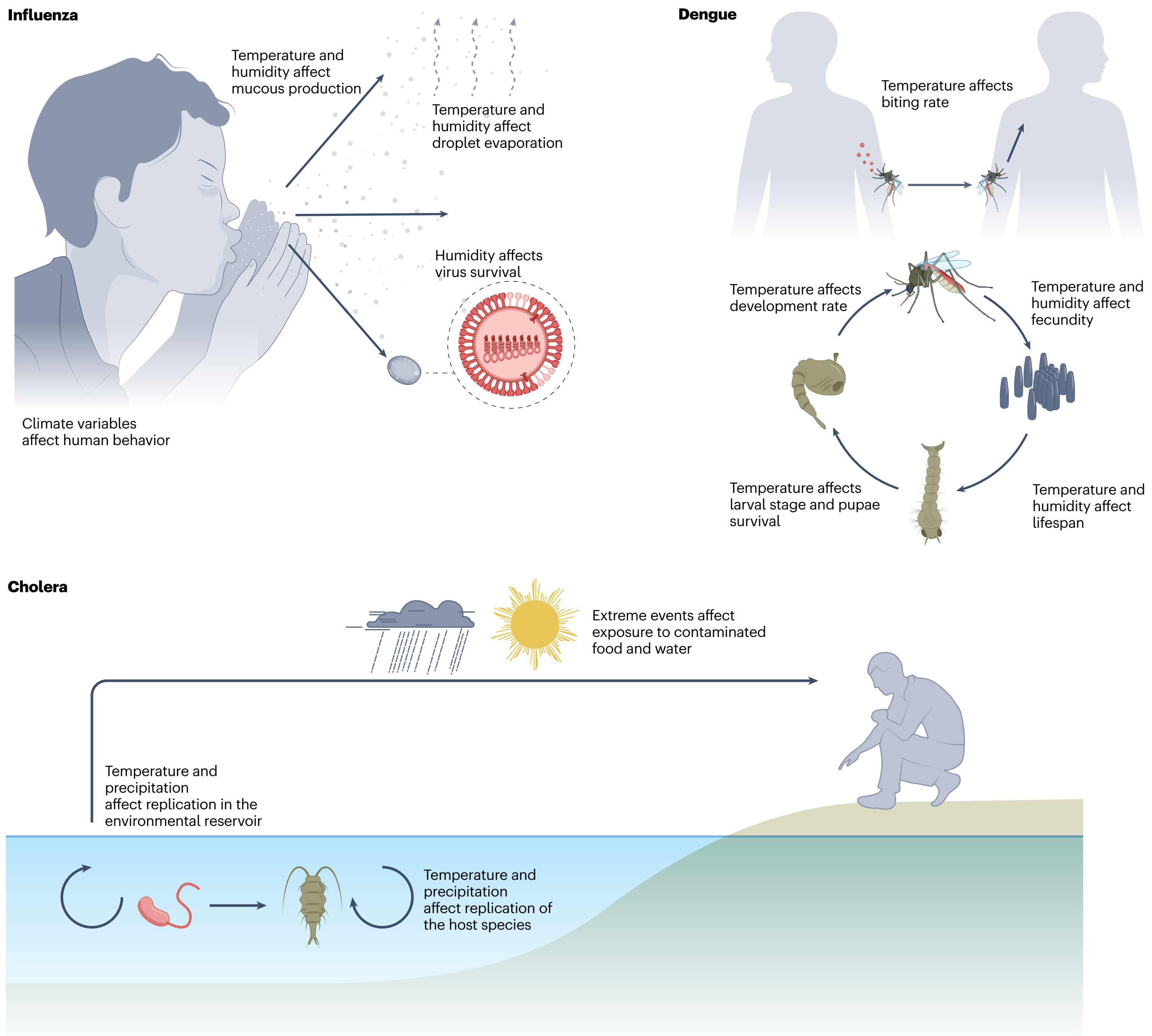

Fig. 1 | The role of climate drivers on transmission. Pathways are shown for a respiratory pathogen (influenza), a mosquito-borne pathogen (dengue) and a disease with an environmental reservoir (cholera).

The impacts of climate on environmentally transmitted pathogens are pathogen specific and context dependent[43,44]. Cholera offers a textbook example of the complex relationship between waterborne disease transmission and climate[45,46], whereby precipitation impacts cholera transmission via several context-dependent mechanisms. Increased rainfall partially triggered the 2015 South Sudan cholera outbreak, likely owing to increased exposure to already contaminated water rather than increased water contamination[47]. In contrast, a 2019 outbreak in Uganda was directly linked to well contamination following heavy rainfall[48]. Indirect climate impacts on the *Vibrio cholerae* reservoir are mediated through the commensal relationship between the bacteria and plankton, especially copepods (tiny crustaceans that are a major component of plankton ecosystems)[49]. The concentration of *V. cholerae* in an infected copepod is high enough to cause cholera in humans upon consumption, meaning climate-driven copepod population changes can influence cholera dynamics[50]. For example, oceanic algal blooms, driven by changes in sea surface temperatures and precipitation, are associated with proliferation of *Vibrio*-containing copepods, which may partially explain increases in cholera incidence[51]. While the influence of El Niño (a climate pattern occurring every 2–7 years) on cholera has long been recognized as important[52,53], newer evidence suggests that temperature anomalies may interact with the emergence and replacement of cholera strains, creating conditions that favor the spread of novel variants[54]. Broadly, extreme events (for example, flooding and drought) associated with climate change are expected to increase cholera's reproductive number, although this effect is highly contingent on socioeconomic factors, particularly sanitation, poverty, conflict and healthcare[55].

Climate variables affect seasonal outbreak patterns of infectious disease

A footprint of the impact of climate on infectious diseases is the ubiquity of seasonal fluctuations in incidence. However, seasonal patterns of outbreak may be driven by other (non-climate) seasonally

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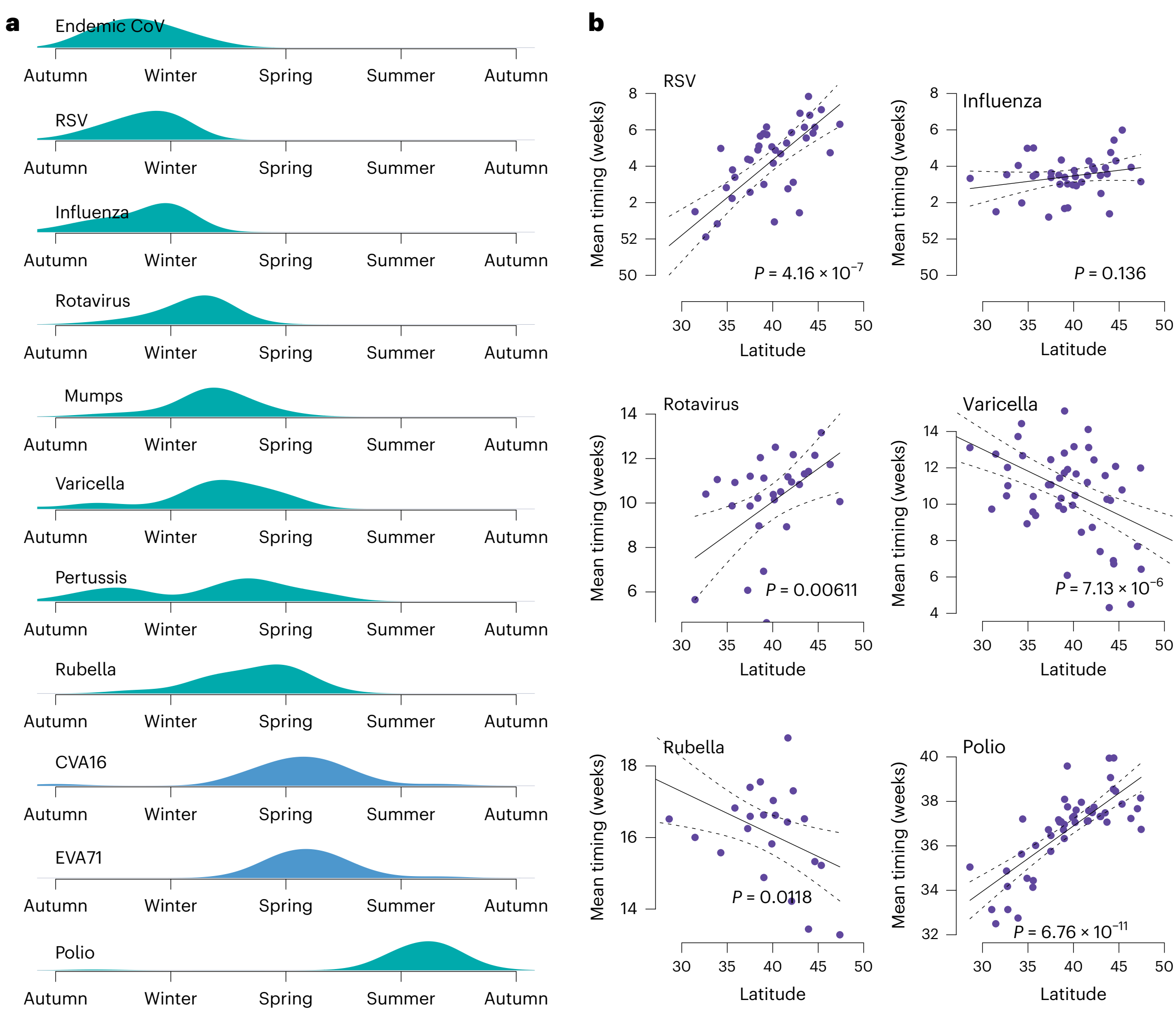

Fig. 2 | The seasonality of directly transmitted diseases. a, The mean timing of the seasonal outbreak peak for 11 directly transmitted diseases based on data from US-based Project Tycho[118] (green) and data from the China Centers for Disease Control[119] (blue). Density plots show timing of peaks relative to the summer months, defined as the warmest month for each state or province. Effective vaccines were introduced for several of these pathogens, reducing circulation, so the figure uses pre-vaccination era data, specifically for rotavirus, mumps, varicella, pertussis, rubella and polio. **b**, Scatterplots of latitude against mean outbreak timing for six diseases using US state-level data from Project Tycho[118]. A fitted linear regression line (black) is shown, along with 95% confidence intervals (dashed lines) and corresponding *P* values for the regression model, indicating the statistical significance of the fit.

varying drivers such as seasonal population mobility[56] or schooling[57,58]. Figure 2a shows the mean peak timing of 11 directly transmitted pathogens (see Fig. 3 for vector-borne diseases) using data from the USA and China. The seasonal timing of directly transmitted disease outbreaks varies from the winter (endemic coronaviruses, RSV and influenza) to the summer months (enteroviruses CVA16 and EVA71, and polio). Some diseases, such as pertussis (a bacterial infection), show a more evenly distributed timing of outbreaks (peaks occurring from autumn through to early summer), which may speak to weaker seasonal forcing. However, many pathogens exhibit a clear seasonal outbreak period; for example, RSV occurs during the autumn to early winter period, peaking just before the influenza season.

The differential timing of outbreaks across pathogen species implies a lack of a common driver, or different environmental sensitivity. For example, increased time spent indoors during the winter months cannot simultaneously explain the timing of influenza (winter) and polio (summer/autumn). However, there are some coarse groupings across seasonality by pathogen: enveloped viruses (that is, viruses with an outer lipid layer, including coronaviruses, RSV, influenza and the viruses that cause mumps, varicella and rubella) appear more likely to have winter outbreaks, whereas non-enveloped viruses (enteroviruses CVA16, EVA71, poliovirus and rotavirus) are more likely to have spring or summertime outbreaks[59]. Experimental work suggests that lower temperatures lead to increased ordering of lipids in the viral envelope, increasing stability[60,61]. Further work is required to understand the mechanistic link between virus structure, transmission pathway and climate variables across pathogen types. Understanding these links could improve our ability to project the possible effects of climate change on the seasonality of directly transmitted diseases.

Seasonal outbreak patterns may be driven by climate variables or other factors such as human behavior[62]. A striking indication of a link between climate variables and transmission is when the timing, intensity or periodicity of seasonal outbreak patterns varies consistently over space, for example, latitude (Fig. 2b). Certain diseases, for instance polio[59,63,64] and RSV[30,65], exhibit strong latitudinal gradients in mean timing with later epidemics at higher latitudes. Differences in periodicity of RSV outbreaks over space have been noted: northern counties in the USA are more likely to exhibit biennial cycles (that is, a large outbreak every other year), whereas southern and tropical locations exhibit outbreaks year-round, sometimes with two smaller epidemic peaks a year[30].

However, not all diseases that are expected to be climate-driven experience clear latitudinal gradients; influenza, for instance, exhibits strong synchronicity at the state level in the USA[66], but more striking

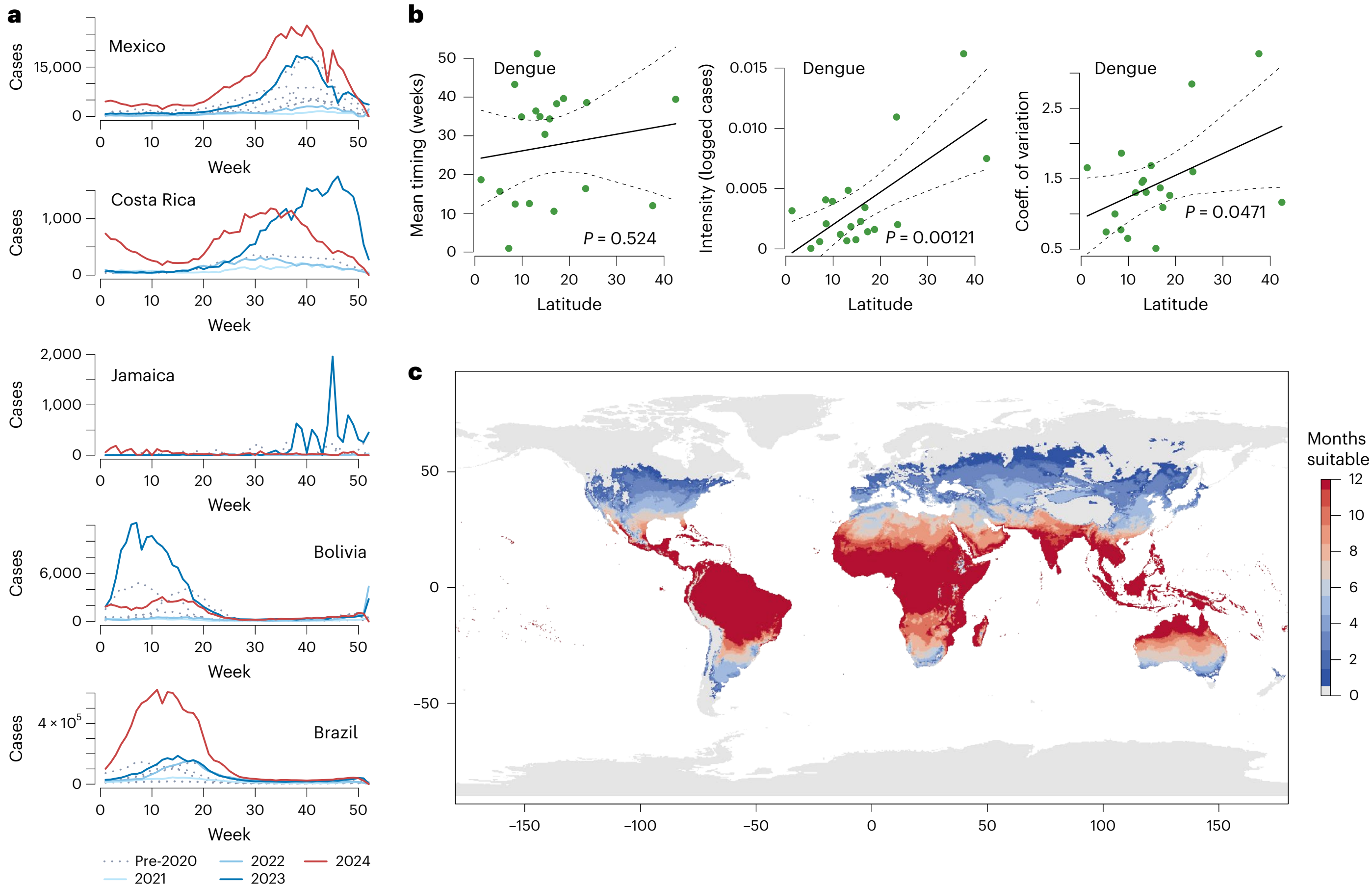

Fig. 3 | Geographic range and extreme outbreaks of vector-borne diseases. **a**, Seasonal outbreak patterns of dengue in the Americas over the last 5 years based on data from OpenDengue. **b**, Scatterplots of latitude against dengue timing, intensity and coefficient of variation of dengue cases using data from the Americas. A fitted linear regression line (black) is shown, along with 95% confidence intervals (dashed lines) and corresponding *P* values for the regression model, indicating the statistical significance of the fit. **c**, Estimated months suitable for dengue transmission are based on thermal ranges from ref. 10 and ERA5 climate reanalysis data[120].

gradients at the national level[67,68]. At the subnational scale, regional and local mobility may have a stronger role than climate in determining the outbreak timing for diseases like influenza[66,69], and changes in patterns of susceptibility driven by schooling can alter the timing of seasonal outbreaks even when climate drivers remain relatively unchanged, illustrated by RSV seasonality in Japan[70]. A better synthesis of behavioral and environmental factors into mechanistic disease models would help elucidate the implications of these effects. Improved integration of statistical inference approaches with mechanistic models may help disentangle the effects of different drivers[71].

Vector-borne diseases and diseases with an environmental reservoir also exhibit seasonal outbreak patterns that may be modulated by climate variables[58] (Fig. 3). An important distinction is that many directly transmitted diseases are globally distributed. For instance, human influenza is found on all continents inhabited by humans[72]; changes to seasonality are one of the key indicators for a role for climate in this case. In contrast, many vector-borne diseases and diseases with an environmental reservoir only cause outbreaks in certain locations[17,73]. In this case, the distribution of the pathogen may provide clues as to the role of climate; the geographic range of the disease may reflect the thermal tolerance of the host[10], for example, or the suitability of the environmental reservoir[74]. While latitudinal gradients, seasonal outbreaks or the existence of a thermally limited geographic distribution are indicative of a role for climate variables, understanding the climate–transmission link often requires detailed modeling and experimental work. For example, modeling the nonlinear effects of temperature and precipitation on cholera transmission implicated summer heat and monsoon rains in biannual outbreaks observed in Kolkata[75]. Once the linkage between climate and transmission is well understood, models can be leveraged to understand how disease trajectories may evolve in the future as the climate changes.

Climate change affects projected and historical disease trajectories

There are many studies linking environmental variables to population-level outbreak patterns in the present[42]. If climate variables influence the transmission of a particular pathogen, it is expected that projected changes in the climate may affect the seasonality or geographic distribution of the disease in the future. However, developing quantitative projections for the future effects of climate change on infectious disease requires integrating current understanding of the climate–transmission relationship with climate projections that are output from general circulation models based on possible future scenarios, known as Shared Socioeconomic Pathways (SSPs; Fig. 4). Quantitative projections of the effects of climate change on disease outcomes remain rare; however, these projections are valuable in allowing public health planners to allocate resources to mitigate emerging threats.

In terms of quantitative projections, vector-borne diseases, specifically mosquito-borne diseases, have received the most attention. Mechanistic understanding of the link between climate variables and the mosquito life cycle (Fig. 1) has been used to understand the potential geographic range (Fig. 3c) of these diseases in the present, and

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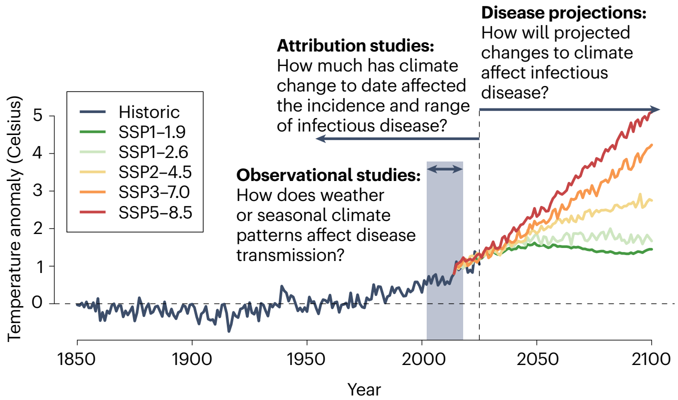

Fig. 4 | Timeline of attribution and projection studies. Lines show observed[121] (black line) and projected[122] (colored lines) global temperature anomalies relative to the 1901–2000 average. The gray box shows a hypothetical data range for an observational climate–disease study. Understanding the effect of climate change on infectious disease requires combining estimates based on short-term observational data with either future climate projections, or longer-term historical observations (attribution studies). Rarely, health datasets have the temporal range for the historical effect of climate change on a health outcome to be observed directly[123].

how this range might change in the future under climate change[17,73,76]. Recent work by Childs et al.[5] used a primarily statistical approach to estimate the effect of temperature on dengue incidence, and to project that future warming could increase dengue incidence by 49–76% by 2050. The authors also used their estimates to study how the warming experienced to date could have contributed to the recent expansion in dengue incidence, and found that 18% of dengue incidence in study countries is attributable to climate change. Attribution studies such as these are a nascent area in infectious disease research, and climate–health research more broadly[77] (Fig. 4)—but have a much longer history in climate science, specifically in the attribution of extreme weather events. A recent review found only 13 studies attributing health impacts to climate change[78]. More work is needed to understand the extent to which climate change has already impacted human health over the last century.

For directly transmitted diseases, there has been some work to consider the projected effect of climate change on the incidence of enteroviruses[59], RSV[30] and influenza[79]. However, this work remains geographically limited and to date no global projections exist. An emerging theme of this work is that climate change will likely alter the seasonal outbreak patterns of directly transmitted diseases, but may not necessarily affect the total burden: given the global distribution of these diseases, population immunity may be high enough to limit large increases in total incidence[80]. This differs from the finding for vector-borne diseases, where changes to the climate may drive the disease to expand into areas with minimal prior outbreaks—and hence lower population immunity[5]. Among diseases with an environmental reservoir, projections suggest that environmental suitability for coccidioidomycosis, the fungal pathogen that causes valley fever, will expand with climate change[74]. For waterborne diarrheal diseases including cholera, climate projections imply an increase in burden despite recent improvements in sanitation, although the projected effects vary by pathogen[81].

The risk from climate variability and extremes

Extreme weather events—such as extreme rainfall or heatwaves—may facilitate disease spread through enhancing or extending conditions favorable for pathogen transmission. Changes in precipitation can exert multifaceted impacts on vector-borne disease outbreaks, for example by generating new mosquito breeding grounds. Ugandan villages bordering a flood-affected river face a 30% higher malaria risk compared with villages farther away[82]. Similarly, the World Health Organization reported a fourfold increase in malaria cases in Pakistan following the torrential floods in 2022 compared to the previous year[83]. Despite this, drought conditions do not consistently reduce vector-borne disease risk; while drier conditions can eliminate some mosquito breeding habitats as naturally occurring standing water dries[84], they can also alter human water-use behaviors, encouraging storage practices that may create new breeding grounds. Consistent with this, Lowe et al.[85] observed higher dengue burden following extreme droughts in areas with frequent water shortages, where stored water containers may act as vector breeding sites, compared to those with more reliable water access. Respiratory viruses are likewise sensitive to climate extremes: tropical regions may be especially vulnerable to climate shocks, with both high and low ranges of specific humidity associated with elevated influenza transmission[86].

Climate change may interact with natural climate phenomena such as El Niño Southern Oscillation (ENSO), to exacerbate risk[87]. A recent example is the global surge in dengue in 2024 in which over 14 million cases were reported globally (in comparison to 7 million in 2023) (refs. 6,88) (Fig. 3a). In Brazil, one of the countries with the highest dengue burden at this time, the vector *Aedes aegypti* expanded in range from 2015 onwards, reaching higher altitudes and more southern provinces, a trajectory likely attributable to warming temperatures[89–91]. In 2024, El Niño amplified rainfall and flooding in parts of Brazil, providing ample opportunity for mosquito breeding[92]. At the same time, the country detected all four dengue serotypes, with DENV-3 having recently reemerged[93]. It is likely that a combination of warm temperatures driven by climate change, rainfall driven by ENSO and population susceptibility to a reemerging serotype created suitable conditions for a large outbreak. Other South American countries like Colombia were similarly impacted by ENSO and observed outbreaks caused by multiple serotypes, suggesting a role for climatic conditions rather than features of a specific viral lineage[94]. ENSO also facilitated favorable conditions for dengue in other parts of the world, such as Indonesia[95]. Many countries in Europe experienced outbreaks of their own, driven in part by warming summers and contributing to the global surge in dengue[96]. Such pathogen emergence at higher latitudes may be associated with variably sized (and potentially large) outbreaks occurring at unexpected times of year, owing to higher susceptibility (Fig. 3b).

Together, these studies suggest that extreme weather events and climate variability may trigger critical tipping points for disease dynamics, amplifying risks for both vector-borne diseases and respiratory pathogens, and underscoring the need to account for climate variability alongside long-term climate shifts in public health planning. While early warning systems have traditionally been used to reduce risk to humans during hazardous events like hurricanes or volcanic eruptions, there is growing interest in adapting these approaches to track the likelihood of disease outbreaks in particular locations[97]. In Madagascar in 2022 and 2023, major tropical cyclones disrupted routine malaria control activities (for example, malaria chemoprophylaxis and treatment) and, consequently, malaria infection rates were high among school-aged children in affected areas in the months following the cyclones[98]. This provides an interesting test case, highlighting the need for climate-informed, proactive interventions. Toward this goal, Rice et al.[99] used mathematical modeling to show that targeted vaccination deployment—a recently available tool for malaria control—has the potential to prevent approximately half of all symptomatic infections during storm aftermath if sufficient coverage (for example, >70%) is reached in cyclone-prone areas. Robust early warning systems could prompt timely responses before extreme weather events, such as launching vaccination campaigns or allocating resources to local healthcare systems, to minimize illness and prevent healthcare system strain.

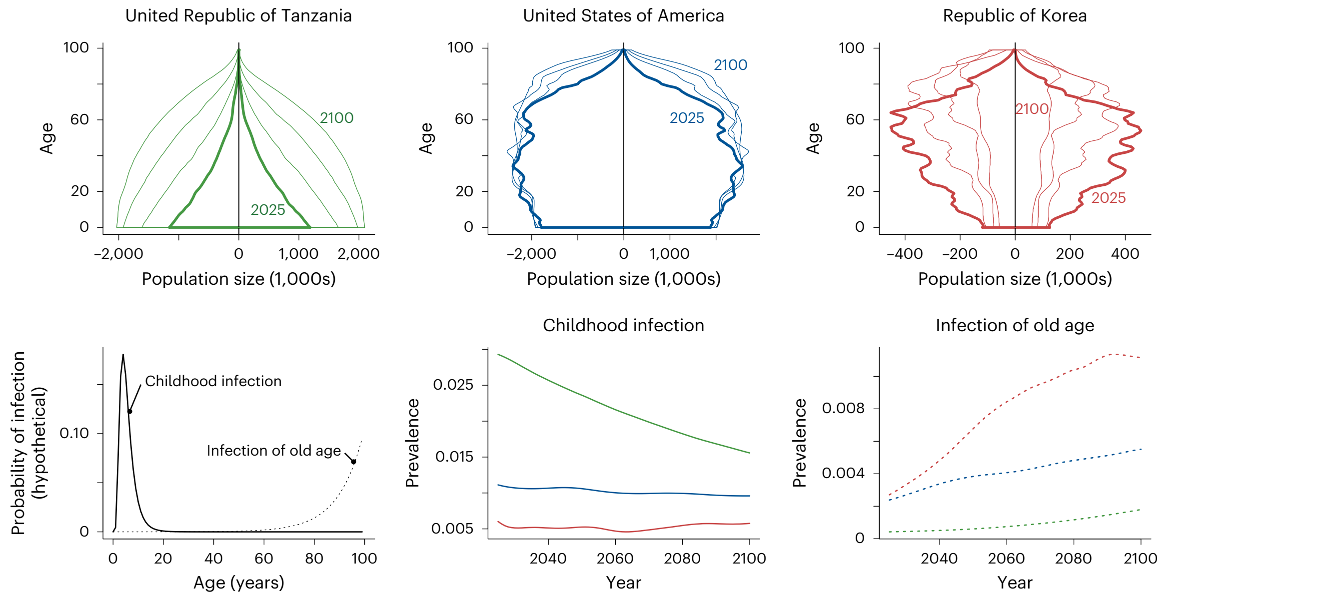

Fig. 5 | Demography and the burden of infection across age. United Nations-projected age structure[124] reveals aging populations through time (years 2025, 2050, 2075 and 2100 shown with declining line widths, and via labels), illustrated across exemplar countries (top row). Both the underlying structure and the trajectory of aging will add an additional perturbation to the burden of climate-sensitive infections, here illustrating the demographic impact on the trajectory for an infection with greatest impact on the young (for example, diphtheria) or an infection most affecting the very old (for example, coronavirus). Infection probabilities for both are shown on the bottom left; the consequent prevalence of the former declines with aging populations (bottom middle); while the latter increases (bottom right); but in each case the background demographic structure (colors) affects burden in predictable ways.

Accounting for demographic change

Climate change is occurring at a time of unprecedented demographic change. Globally, declines in fertility rates are leading to aging populations (Fig. 5) and household size has fallen precipitously[100]. Additionally, since 2008, over half the population of the planet lives in cities[101]. These demographic changes may independently alter the risk from infectious disease, depending on the age profile associated with infection (Fig. 5), or as a direct consequence of how urbanization shapes patterns of risk[1] (for example, modifying exposure to vectors)[102]. Declining birth rates may also reduce the supply of susceptible individuals and thus infection dynamics for pathogens that are immunizing[57].

Demographic changes may also interact with climate change. As one example, the location of urban centers may shift the global burden of infection by altering which infections thrive under the climate conditions where most people live. This could alter patterns of pathogen evolution, by modifying exposure to climate-driven bottlenecks in transmission[103]. Subtle interactions between demography and climate may also manifest: for immunizing infections, declining birth rates generally increase the average age of infection[104], which might be counterweighted or amplified by changes in transmission associated with climate change, as increases in transmission generally reduce the average age of infection. Finally, climate change itself can drive demographic changes in migration[105]. This has the potential to modulate infectious disease risk via accelerated movement of pathogens to clusters of susceptible people, increasing their risk[106].

In some instances, demographic change may have considerably larger impacts than climate change. For example, for toxoplasmosis, a teratogenic infection with an environmental transmission pathway, rapid global increases in the age at fertility may be affecting the burden of disease more swiftly than the increase in temperature or other climate variables[107]. Similarly, a recent modeling exercise indicates that aging populations may do more to amplify the burden of chikungunya virus than climate-driven expansions in vector range[108]. However, disentangling these varied impacts requires careful analysis, and much more research is needed.

Future directions

The projected effects of climate change on infectious disease outbreaks are expected to be strongly pathogen dependent, heterogeneous over space and nonlinear. Nonlinearities can arise from the shape of relationships between climate drivers and transmission (Fig. 6a,b), as well as the underlying dynamics of the disease system (Fig. 6c). Such nonlinearities may make the effects of climate extremes particularly hard to anticipate. In order to prepare, new tools may be required that leverage the proliferation of available climate and disease data streams, while developing new data pipelines, to predict possible changes to the timing and intensity of outbreaks. Predictive tools should be coupled with appropriate policy tools, including control measures that can be implemented in advance of an outbreak to mitigate risk. Below we consider some key areas for future work.

New data streams

Since the COVID-19 pandemic, there have been increasing efforts to improve surveillance of infectious diseases and increase data availability. However, surveillance for cases of disease (ranging from passive syndromic surveillance to active clinical case detection) has limitations, including different reporting approaches that vary by location. Wastewater surveillance provides a promising new data stream that may overcome certain issues associated with clinical surveillance[109], improving speed and, for some pathogens, providing an estimate of prevalence. However, case surveillance data (either at the individual level, or via wastewater) do not track the size of the susceptible pool, which is important for determining risk (Fig. 6c). During the early years of the COVID-19 pandemic, serological surveys were increasingly used to track immunity; however, these efforts have not been sustained in recent years[110]. It has been proposed that a Global Immunological Observatory could pool serological information from multiple locations to address this need[111,112], yet such an effort would require a multinational coordination. If privacy challenges can be met, expanding surveillance for cases of disease to encompass the landscape of immunity could be paired with datasets on human mobility (for example, mobile phone

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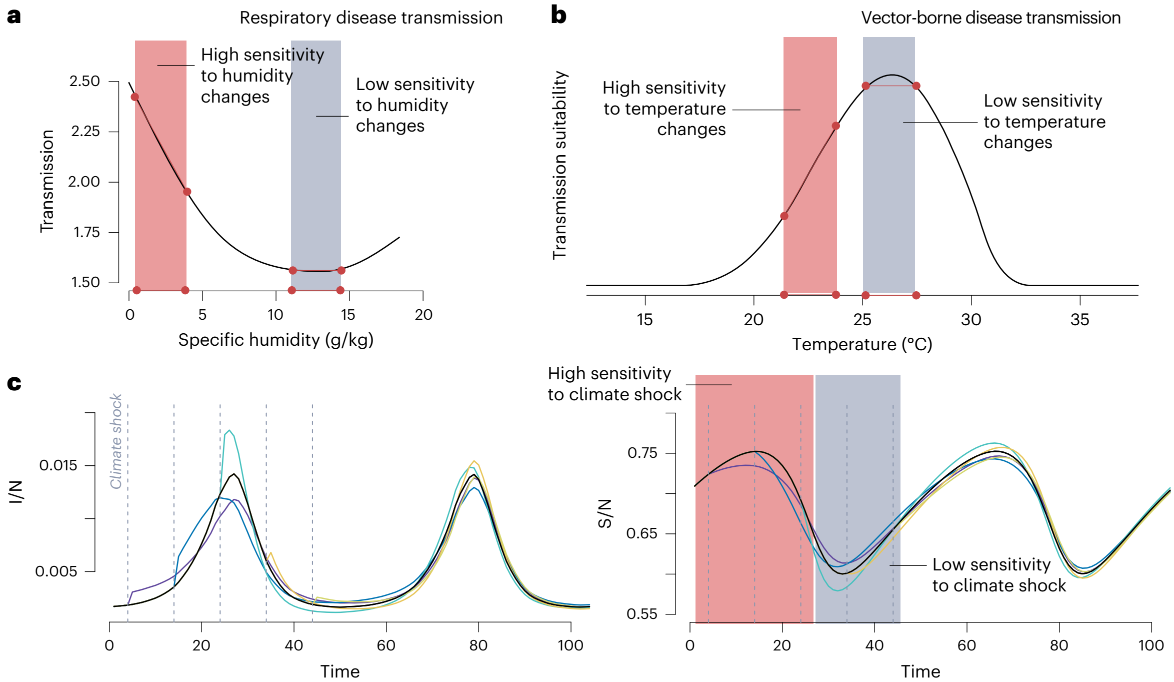

Fig. 6 | Illustration of nonlinear effects of climate on disease dynamics. **a**,**b**, Many pathogens, for example respiratory (**a**) and vector-borne (**b**), exhibit a nonlinear relationship between climate and transmission. The same increase in climate will thus have very different effects depending on the baseline, either radically increasing transmission (red shaded area), or having minimal effect (gray area). **c**, The same climate shock (colored lines) can also have different effects on the proportion infected (I/N) depending on the underlying susceptible (S/N) population size, as transmission requires new susceptibles. Shocks closer to peak susceptibility (right) may have a greater impact on outbreak size (left), than the same shock when the proportion susceptible is low. The graph is based on simulations that were run using a model for influenza[86].

call data records[113]) to direct health sensors (for example, monitoring temperature, heart rate[114]) to sharpen our understanding of disease risk.

Artificial intelligence

With the increasing availability of large climate and disease datasets—for example, remote-sensing and reanalysis datasets for climate and electronic health record information—there is a clear opportunity to develop machine learning models that synthesize these data to develop infectious disease forecasts and early warning systems. Early warning systems can leverage climate forecasts to predict climate-sensitive disease outbreaks[115], and machine learning models can provide an array of approaches to develop these predictions[116]. However, a concern for machine learning approaches is whether they can accurately recover the underlying mechanics of disease transmission in order to be valuable under extreme scenarios where changes to susceptibility may amplify risk. Physics-informed neural networks may be leveraged in this case to improve prediction while accounting for the known theory of disease transmission[117]; however, more work is needed to develop this toolkit such that it can be applied in a public health setting.

Health systems

As climate change alters the context for infectious disease transmission, health systems must be ready to respond. Identifying and reinforcing supply chains of drugs or other essential medical supplies that might be at risk in a context of a changing climate is one important step; another is evaluating the portfolio of interventions available and their robustness in the context of climate disruptions—for example, use of vaccines to provide lasting protection in areas at risk of disruptions such as cyclones[99]. A key theme of recent work on climate change and infectious disease outbreaks is to expect the unexpected: pathogens that do not typically circulate in a location may begin to circulate as the climate warms; locations with wintertime disease outbreaks may begin to see summertime outbreaks if the winter is less suitable for transmission. Doctors and public health practitioners must remain vigilant to these emerging risks; flexible public health response measures that can be adapted to a changing context should be developed.

Conclusion

Climate change will alter future patterns of infectious disease, and this will occur against the backdrop of a rapidly aging population that is more geographically connected than ever before[1]. A growing area for climate–disease work is to consider how climatic, demographic and behavioral change may jointly reshape infectious disease burden in the future. This should be coupled with comprehensive work on weather extremes, which may be exacerbated by climate change, or occur against a shifting landscape of risk. For pathogens where there is a coalescing understanding of the effect of mean changes, new directions could consider the coupled effect of mean changes and extremes; for instance, how dengue burden may be affected by El Niño in a warming world.

Most of the pathogens discussed above exhibit a nonlinear relationship between climate variables and transmission. It is important to note, therefore, that qualitatively, climate change may cause some regions to experience increasing risk, while others experience decreasing risk—and that these risks will vary substantially by pathogen and location. More work is needed to understand this heterogeneity. Effective interventions will require detailed, location-specific projections that can be used to allocate resources across areas of possible burden, building resilience where it is most needed.

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All authors: writing original draft, reviewing and editing; R.E.B. and C.J.E.M.: data analysis and visualization.

Competing interests

The authors declare no competing interests.

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